

Op-Amp-Based Pulse-Width Modulation for Energy-Efficient Sensor-Controlled Lighting

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Abstract—This project presents an analog pulse-width modulation (PWM) controller for energy-efficient lighting. The circuit uses four UA741 general-purpose operational amplifiers to generate a triangular carrier, sense ambient light through a light-dependent resistor (LDR), and compare the two signals to produce a PWM waveform. The PWM output drives an N-channel MOSFET, which switches an LED without forcing the op-amp to supply the load current directly. In bright conditions the LDR raises the control voltage and reduces the LED duty cycle; in darkness the control voltage falls and the duty cycle increases. For the selected timing components, the calculated switching frequency is approximately 133 Hz. The design therefore demonstrates a compact, low-cost lighting controller that changes average LED power automatically and does not require a microcontroller or programming.

Index Terms—pulse-width modulation, operational amplifier, LDR, automatic lighting, UA741, MOSFET, energy efficiency

I. INTRODUCTION

Pulse-width modulation controls the average power delivered to a load by changing the fraction of each switching period for which the load is on. This fraction is the duty cycle,

$$D = \frac{T_{on}}{T} \times 100\%, \quad (1)$$

where T_{on} is the high-state time and T is the switching period. A common analog method compares a slowly varying control voltage with a triangular carrier. The comparison changes pulse width while leaving the carrier frequency nearly constant [1], [2].

The objective of this project was to design and build an op-amp PWM circuit that automatically adjusts an LED according to ambient light. The intended response is intuitive: the lamp becomes dim or turns off when sufficient light is available and becomes brighter as the surroundings become darker. This reduces unnecessary energy use and can be applied to corridor lamps, night lights, garden lighting, and small solar-lighting systems.

II. IMPLEMENTATION

A. System Operation

The controller consists of four functional sections. First, a Schmitt-trigger comparator produces a square wave. Second, an integrator converts the square wave into a triangular carrier.

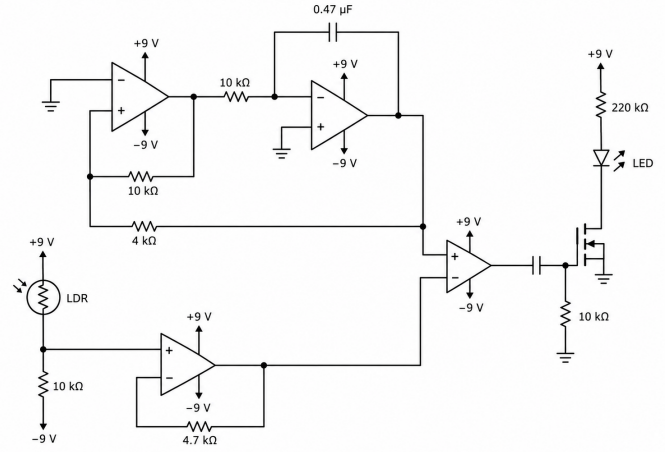


Fig. 1. Complete op-amp PWM lighting controller. U1 and U2 form the carrier generator, U3 buffers the LDR signal, and U4 creates the PWM drive. U1–U4 are four UA741 operational amplifiers.

Third, an LDR voltage divider and buffer produce a light-dependent control voltage. Finally, a comparator produces PWM and a 2N7000 MOSFET switches the LED. Four separate UA741 op-amp ICs (U1–U4) are used, one per functional stage, since the UA741 is a single general-purpose operational amplifier rather than a quad package [3]. Because the UA741’s common-mode input range and output swing do not approach the negative supply rail as closely as a single-supply device would, every stage is referenced around the mid-supply voltage V_{REF} (4.5 V) generated by the reference divider, which keeps all signals within the UA741’s linear operating range.

B. Carrier Generator Design

U1 operates as a Schmitt trigger and U2 as an inverting integrator. The integrator changes direction whenever the triangular signal reaches a Schmitt threshold. Integration of a nearly constant square-wave level produces an approximately linear ramp, which is the standard basis of an analog PWM carrier generator [4].

For a first-order estimate, the threshold ratio is

$$\beta = \frac{R_T}{R_H} = \frac{4 \text{ k}\Omega}{10 \text{ k}\Omega} = 0.40. \quad (2)$$

Assuming approximately symmetrical output swing around the 4.5 V reference, the oscillation frequency is

$$f_{\text{PWM}} \approx \frac{1}{4\beta R_I C_I} = \frac{1}{4(0.40)(10 \text{ k}\Omega)(0.47 \text{ }\mu\text{F})} \approx 133 \text{ Hz}. \quad (3)$$

The actual frequency can differ slightly because the UA741 does not swing equally close to both supply rails and because of resistor and capacitor tolerances.

C. Light Sensor and Duty-Cycle Control

The LDR and a 10 k Ω resistor form a voltage divider. With the LDR connected to 9 V, the sensor voltage is

$$V_{\text{LDR}} = 9 \frac{10 \text{ k}\Omega}{R_{\text{LDR}} + 10 \text{ k}\Omega}. \quad (4)$$

The LDR resistance decreases in bright light, so V_{LDR} rises. U3 is a voltage follower that prevents the oscillator and comparator from loading the divider. U4 compares the triangular carrier at its non-inverting input with the buffered sensor voltage at its inverting input. Consequently, the PWM output is high while $V_{\text{TRI}} > V_{\text{S}}$. If the carrier varies between V_L and V_H , the ideal duty cycle is approximated by

$$D \approx \frac{V_H - V_S}{V_H - V_L} \times 100\%, \quad 0\% \leq D \leq 100\%. \quad (5)$$

Thus bright light produces a small duty cycle, while darkness produces a large duty cycle.

D. MOSFET and LED Driver

The comparator output is connected to the MOSFET gate through a 220 Ω resistor. A 10 k Ω pull-down resistor keeps the LED off if the op-amp output is disconnected. The 2N7000 is an N-channel switching MOSFET suitable for low-voltage, low-current loads [5]. For a red LED with an assumed forward voltage of 2 V and a target current of 15 mA, the series resistance at a 9 V supply is

$$R_{\text{LED}} = \frac{9 - 2}{0.015} \approx 467 \text{ }\Omega. \quad (6)$$

The nearest standard value, 470 Ω , was selected. If the circuit is operated from 5 V, a 220 Ω LED resistor is appropriate for approximately 14 mA. A value of 220 k Ω must not be used because it would limit the current to only a few tens of microamperes.

E. Components and Construction

Table I lists the principal hardware. A 100 nF ceramic bypass capacitor should be placed close to the supply pins of each UA741. The prototype should use short jumper wires and a common ground to reduce switching noise.

TABLE I
MAIN COMPONENTS

Component	Quantity / value
UA741 op-amp	4
LDR	1
2N7000 N-MOSFET	1
LED	1
Resistors	220 Ω , 470 Ω , 4 k Ω , 10 k Ω
Timing capacitor	0.47 μF
Bypass capacitor	100 nF (per IC)
Reference capacitor	10 μF
DC supply	9 V
Breadboard and jumpers	As required

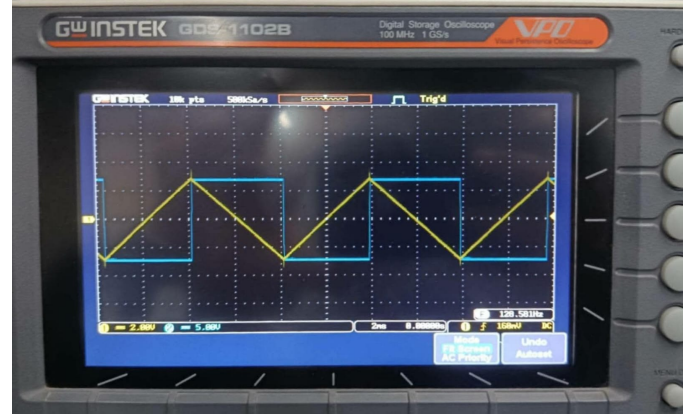


Fig. 2. Simulated triangular carrier and PWM output.

F. Simulation and Experimental Procedure

The circuit can first be simulated in Proteus, Multisim, or LTspice. The triangle waveform, sensor voltage, PWM output, and MOSFET drain waveform should be observed simultaneously. The hardware test then follows four steps:

- 1) verify $V_{\text{REF}} \approx 4.5 \text{ V}$ before inserting the ICs;
- 2) confirm the square and triangular carrier waveforms;
- 3) illuminate and cover the LDR while measuring sensor voltage and duty cycle; and
- 4) connect the LED stage and confirm that brightness increases as ambient light decreases.

G. Expected Results and Discussion

Table II gives calculated behavior for representative LDR resistances. For illustration, the triangular carrier is taken to vary from approximately 3.3 V to 6.3 V. These values should be replaced by measured values after testing.

The calculated trend confirms the intended negative relationship between ambient light and LED power. Unlike linear dimming, PWM keeps the MOSFET mainly in either the on or off state, which reduces power loss in the switching device. The main implementation challenges were stabilizing the virtual reference, preventing sensor noise from causing duty-cycle jitter, and keeping the op-amps from directly driving the LED

TABLE II
CALCULATED LIGHT-CONTROL RESPONSE

Condition	R_{LDR}	V_S	Duty cycle
Bright	2 k Ω	7.50 V	$\approx 0\%$
Medium	10 k Ω	4.50 V	$\approx 60\%$
Dark	100 k Ω	0.82 V	$\approx 100\%$

current. These issues were addressed with reference and supply decoupling, short connections, an LDR buffer, a gate pull-down resistor, and the external MOSFET driver. Component tolerance and the limited output swing of the UA741 devices remain the main sources of difference between theoretical, simulated, and measured waveforms.

III. CONCLUSION

An op-amp-based PWM controller for automatic LED lighting was designed using a Schmitt trigger, integrator, LDR sensor, comparator, and MOSFET switch, built from four UA741 operational amplifiers. The selected timing network gives a calculated carrier frequency of about 133 Hz. The control method reduces the LED duty cycle in bright conditions

and increases it in darkness, providing automatic illumination without a microcontroller. Future improvements could include an adjustable sensitivity control, a higher PWM frequency, hysteresis at the light threshold, a logic-level power MOSFET for larger lamps, and closed-loop measurement of actual illumination.

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